

Homework #3

CS 4342, Fall 2026

100 points total [13.3% of your final grade]

Due: September 28, 2026 by 11:59pm

Delivery: Submit via Canvas

Part 1. Softmax regression [60 points]

In this part, you will implement softmax regression (in `problem1.py`) with stochastic gradient descent in `python3`.

We provide the following files:

- `problem1.py` - You will implement several functions of softmax regression. Do not change the input and the output of the functions.
- `test1.py` - This file includes unit tests. Run this file by typing `'pytest -v test1.py'` in the terminal. No modification is required.

Part 2. Adding CNN and fully connected layers to recognize handwritten digits on PyTorch [40 points]

In this part, you will deal with the [MNIST Database](#). The MNIST Database is a collection of samples of handwritten digits from many people, originally collected by the National Institute of Standards and Technology (NIST), and modified to be more easily analyzed computationally. We will use a tutorial and sample software provided:

- Read and run [a tutorial](#) to be familiar with how to add CNN layers into PyTorch.
 - Download `hw3-part2.ipynb` and run it to be familiar with the code. Currently it contains two fully connected layers with softmax.
 - Your job is to add one CNN layer with one pooling layer before the two fully connected layers with softmax. Refer to the detailed instruction about the CNN layer. The main difference between the tutorial and this given `hw3_part2.ipynb` is the input image of `hw3_part2.ipynb` has only 1 channel (i.e., gray scale). **Report results of two fully connected layers without the CNN layer and with the CNN layer.**
 - Then, experiment with **at least three additional model configurations**. Each configuration should modify the network architecture and/or training hyperparameters (e.g., # of CNN layers, # of fully-connected layers, # of filters or hidden units, activation functions, learning rate, batch size, # of training epochs, pooling methods). **For each configuration, report the architecture, hyperparameter settings, test accuracy.**
 - Overall, through the experiment, **what is the best configuration? What best prediction accuracy on the test set you got? What did you learn?**
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What to turn in:

- Submit to Canvas your `problem1.py` and pdf document for part 2.
- This is an individual assignment, but you may discuss general strategies and approaches with other members of the class (refer to the syllabus for details of the homework collaboration policy).